Systems Engineering | NYCU 114

Final Term Project Specification

AI-Assisted MBSE Transformation of the Mid-Term Project

Revised Edition · Prepared for Student Groups

Core framing: Use AI to transform, audit, connect, and challenge your system model—not simply to rewrite your mid-term report.

1. Project Purpose

Each group will transform its mid-term systems engineering project into an AI-assisted MBSE package. The final project should demonstrate a connected digital thread:

Stakeholder Needs → ConOps → Operational Requirements → System Requirements → Functions + Architecture → Interfaces → Verification → Traceability → Change Impact

2. Rules and Expectations

- AI may support extraction, structuring, consistency checking, traceability, and simulation design. Students remain responsible for all engineering decisions.
- Do not allow AI to invent facts, standards, thresholds, stakeholder decisions, or performance targets. Mark uncertain items as [ASSUMED], [INFERRED], [NEEDS SOURCE], or [HUMAN DECISION REQUIRED].
- Reward is based on model clarity and traceability, not report length.
- Every major requirement, function, interface, verification case, and change impact should be traceable to a source or rationale.

3. Required Tasks in Order — 8 Tasks

Complete the tasks in sequence. Each task builds on the previous one. The MBSE views they cover are shown in the right column.

#	Task Name	What to Produce	MBSE Views Covered	Replaces / Condensed From
1	**Freeze the Mid-Term Baseline** (Quick anchor — do not over-write)	Summarize original mission, boundary, stakeholders, needs, requirements, architecture. Identify 3–5 weaknesses in the mid-term SE work that MBSE	Baseline	Task 1

#	Task Name	What to Produce	MBSE Views Covered	Replaces / Condensed From
		should fix.		
2	**Build the Operational View**	Write ConOps (mission, users, environment, concept of use). Write 2–3 use cases and one normal scenario + one abnormal scenario. State 3–5 operational success criteria (measurable).	Operational	Task 2
3	**Build the Requirement View**	Convert stakeholder needs into Operational Requirements (OR) and System Requirements (SRS). Each requirement needs: unique ID, rationale, assumption flag, and verification method. Note: parent-child links (OR→SRS) will be captured in the Task 7 RTM—no separate trace column needed here.	Requirements	Task 3
4	**Build the Functional & Architecture View** **[Merged task]**	List the top 5–8 system functions (what the system does, not how). Define major system blocks (subsystems or components). Allocate each function to one or more blocks in a simple table. A standalone function hierarchy is not required—just the allocation.	Functional + Architecture	Merges Tasks 4 + 5
5	**Build the Interface View** **[Scoped: 3–5 interfaces]**	Select the 3–5 most critical external or internal interfaces. For each interface, provide: Interface ID, Sender, Receiver, Data/Material Exchanged, and Linked Requirement. Trigger and failure mode columns are optional (include only if relevant to your system's risk).	Interface	Task 6
6	**Build the Verification View**	Define one verification case per OR (test, analysis, inspection, or	Verification	Task 7 (lightweight)

#	Task Name	What to Produce	MBSE Views Covered	Replaces / Condensed From
		demonstration). Each case needs: Case ID, linked SRS ID, method, pass/fail criterion, and who verifies. Depth over breadth: one well-written case is better than five vague ones.		
7	**Build the Traceability View + Run the AI Audit** **[Merged task]**	**Part A — Traceability RTM:** Build an RTM showing at least one complete digital thread from stakeholder need → OR → SRS → function → block → interface → verification case. Full-matrix coverage is not required; demonstrate the chain for your top 2–3 requirements. **Part B — AI Audit:** Use an LLM to audit the RTM for: missing links, weak requirement language, hidden assumptions, contradictory requirements, and interface gaps. Document what AI found, what you accepted, what you rejected, and why.	Traceability + AI Review	Tasks 8 + 9
8	**Scope One Change Impact** **[Focused, not full rescan]**	Select one meaningful change request (a new stakeholder need, a tighter performance target, or a new constraint). Trace its impact through exactly three layers: (a) Which requirements change or are added? (b) Which blocks or interfaces are affected? (c) Which verification cases must be updated? State the risk introduced by the change. Full re-analysis of all views is not required.	Change Impact	Task 10 (scoped to 3-layer trace, not full rescan)

The final presentation (Task 9 in the original spec) remains a required deliverable — see Section 5.

4. Required Deliverables

Deliverable	Notes
Group MBSE Transformation Report	One integrated report covering all 8 tasks. Does not need to be long—a well-connected model of modest depth is better than a large document with shallow connections.
RTM / Traceability Matrix	Show at least one complete digital thread from need to verification. Partial RTM covering 2–3 top requirements is acceptable; full-matrix coverage is not required.
AI Usage Log	Document prompts used, output accepted, output rejected, human corrections made, and AI risks discovered. Format is flexible—a table or annotated log both work.
Change Impact Analysis	One meaningful change, traced through three layers: requirements, architecture/interfaces, and verification. One to two pages is sufficient.
Final Group Presentation	Explain the transformation from mid-term to MBSE, what AI helped discover, what AI got wrong, and the final digital thread.
Individual Final Thoughts	"Advice to Future SE Students Entering the AI Era." Strict 500-word limit. Personal, specific, and connected to your project experience.

5. Final Group Presentation

The final presentation should cover four things, in any order that tells a clear story:

- The transformation: what changed from the mid-term report to the MBSE model, and why.
- What AI helped you discover: gaps, inconsistencies, or improvements you would not have found alone.
- What AI got wrong: hallucinated facts, wrong assumptions, weak requirements generated by AI, and how you corrected them.
- The digital thread: walk the audience through one complete chain from a stakeholder need to a verification case.

6. Optional Bonus: AI-Assisted System Simulation

Bonus may be awarded only when simulation supports a real systems engineering decision.

- Simulation must link to a requirement, MOE/MOP, or design trade-off.
- Inputs, assumptions, variables, and output metrics must be clearly stated.
- At least two scenarios or sensitivity cases should be compared.
- The result must lead to an engineering decision, requirement revision, or risk insight.

Acceptable tools: Excel, Python, SimPy, AnyLogic, n8n, MATLAB, or any appropriate tool with instructor approval.

7. Grading Structure

Total final term project grade: 100%. Part I is the group project. Part II is the individual final thoughts.

Note on Part I: The Part I criteria below preserve the requested emphasis of 10/40/20/20. Because these add to 90 raw points, the Part I score is scaled to 80% of the final term project grade: Part I final contribution = raw Part I score ÷ 90 × 80.

Part I — Group AI-Assisted MBSE Transformation: 80%

Criteria	Raw Weight	What to Evaluate	Excellent Performance Looks Like
Transformation Clarity	10	How clearly the group transforms the mid-term project into MBSE.	Original weaknesses are identified; improvements are visible and traceable.
MBSE Structure Completeness	40	Operational, requirement, functional, architecture, interface, verification, and traceability views.	A complete digital thread connects needs to verification. Change impact is focused and traceable.
AI Use Quality	20	Prompt log, human correction, AI risk awareness, assumption control.	Students use AI as an engineering reviewer and correct false or weak outputs.
Final Presentation	20	Clarity, professionalism, storytelling, and ability to explain the model.	The audience can understand the system, the model, and the learning within limited time.

Part II — Individual Final Thoughts: 20%

Each student submits an individual reflection titled: "Advice to Future SE Students Entering the AI Era." Strict 500-word limit. Personal, specific, and connected to the project experience.

Criteria	Weight	What to Evaluate
Personal insight	8%	What the student personally learned about SE, MBSE, and AI.
Connection to project experience	5%	Specific examples from the group project, not generic AI discussion.
Understanding of AI limitations	5%	Clear recognition of AI mistakes, assumptions, hallucination risk, and human responsibility.
Usefulness of advice and writing clarity	2%	Advice is practical, concise, and useful for future SE students.

8. Suggested Reflection Questions for Part II

- What did you misunderstand about systems engineering before this course?
- What did MBSE help you see that your mid-term report did not show clearly?
- What did AI do well during the transformation?
- What did AI do poorly or dangerously?
- Which SE skill becomes more important because AI exists?
- What advice would you give to future SE students entering the AI era?

In the AI era, systems engineers will not be valued because they can generate more documents.

They will be valued because they can define the right system boundary, ask the right questions, connect models across the lifecycle, detect inconsistency, make trade-offs, and govern AI-assisted engineering work.